

# Regression Analysis - STAT 482 - Problem Set 9

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## Problem 1

A router is used to cut locating notches on a circuit board. The vibration level is considered to be an important characteristic of the process. Two factors are thought to affect vibration ( $y$ ): bit size ( $x_1$ ) and cutting speed ( $x_2$ ). The data is available as a csv file posted on Blackboard.

### Part (a)

Provide a definition for an interaction effect.

The interaction effect measures the change in an input effect as the other input changes levels.

### Part (b)

Fit an interaction model using coded variables. Compute the coefficient estimates and the standard error.

### Coded data transformation

```
data = read.csv('hw9-1.csv')

to_binary_coded = function(x)
{
  2*(x-mean(x)) / (max(x)-min(x))
}

data$x1 = to_binary_coded(data$bit.size)
data$x2 = to_binary_coded(data$cutting.speed)
data$y = data$vibration
head(data)
```

```
##   bit.size cutting.speed vibration x1 x2    y
## 1   0.0625           40      18.2 -1 -1  18.2
## 2   0.1250           40      27.2  1 -1  27.2
## 3   0.0625           90      15.9 -1  1  15.9
## 4   0.1250           90      41.0  1  1  41.0
## 5   0.0625           40      18.9 -1 -1  18.9
## 6   0.1250           40      24.0  1 -1  24.0
```

### Estimates

```
x.mod = lm(y ~ x1*x2,data=data) # equiv to y~x1+x2+x1:x2 and y~x1+x2+I(x1*x2)
summary(x.mod)
```

```
##
## Call:
## lm(formula = y ~ x1 * x2, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.975 -1.550 -0.200  1.256  3.625
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  23.8312     0.6112   38.991 5.22e-14 ***
## x1           8.3188     0.6112   13.611 1.17e-08 ***
## x2           3.7687     0.6112    6.166 4.83e-05 ***
## x1:x2        4.3562     0.6112    7.127 1.20e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.445 on 12 degrees of freedom
## Multiple R-squared:  0.9581, Adjusted R-squared:  0.9476
## F-statistic: 91.36 on 3 and 12 DF,  p-value: 1.569e-08
```

```
coef(x.mod)
```

```
## (Intercept)          x1          x2          x1:x2
##    23.83125     8.31875     3.76875     4.35625
```

We see that

$$\hat{E}(Y|x) = 23.83 + 8.32x_1 + 3.77x_2 + 4.36x_1x_2.$$

## Part (c)

Write the estimated regression as a function of  $x_1$  for  $x_2 = 1, 0, -1$ .

We have the regression function  $\hat{E}(Y|x)$ , so, for instance,  $\hat{E}(Y|x_2 = 0) = 23.83 + 8.32x_1 + 3.77(0) + 4.36(0)x_1 = 23.83 + 8.32x_1$ . However, in R, we may compute the slope and intercept estimates with:

```
b0 = coef(x.mod)[1]
b1 = coef(x.mod)[2]
b2 = coef(x.mod)[3]
b12 = coef(x.mod)[4]
reg.estimates.x1 = matrix(c(b0+b2,b0,b0-b2,b1+b12,b1,b1-b12),nrow=3)
dimnames(reg.estimates.x1) = list(c("x2=+1","x2=0","x2=-1"),
                                   c("intercept","slope"))
reg.estimates.x1
```

```
##      intercept      slope
## x2=+1  27.60000  12.67500
## x2=0   23.83125   8.31875
## x2=-1  20.06250   3.96250
```

Thus,

$$\hat{E}(Y|x_1, x_2 = +1) = 27.600 + 12.675x_1,$$

$$\hat{E}(Y|x_1, x_2 = 0) = 23.831 + 8.319x_1,$$

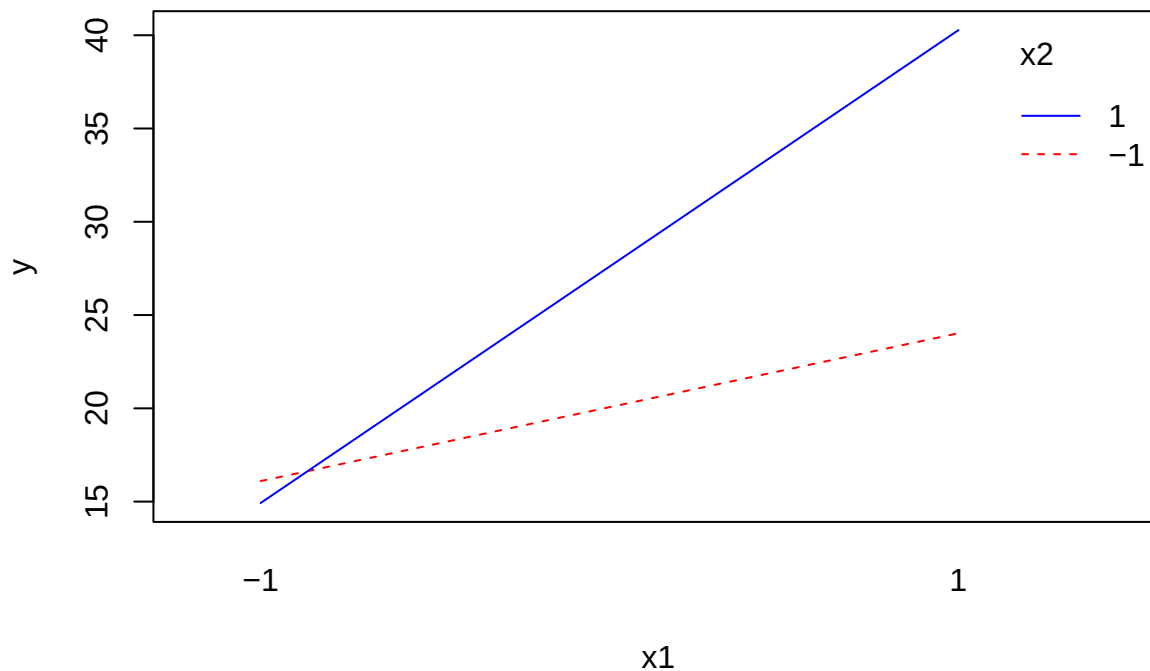
$$\hat{E}(Y|x_1, x_2 = -1) = 20.063 + 3.963x_1.$$

## Part (d)

Create interaction plots for both the interaction model and the additive effects model.

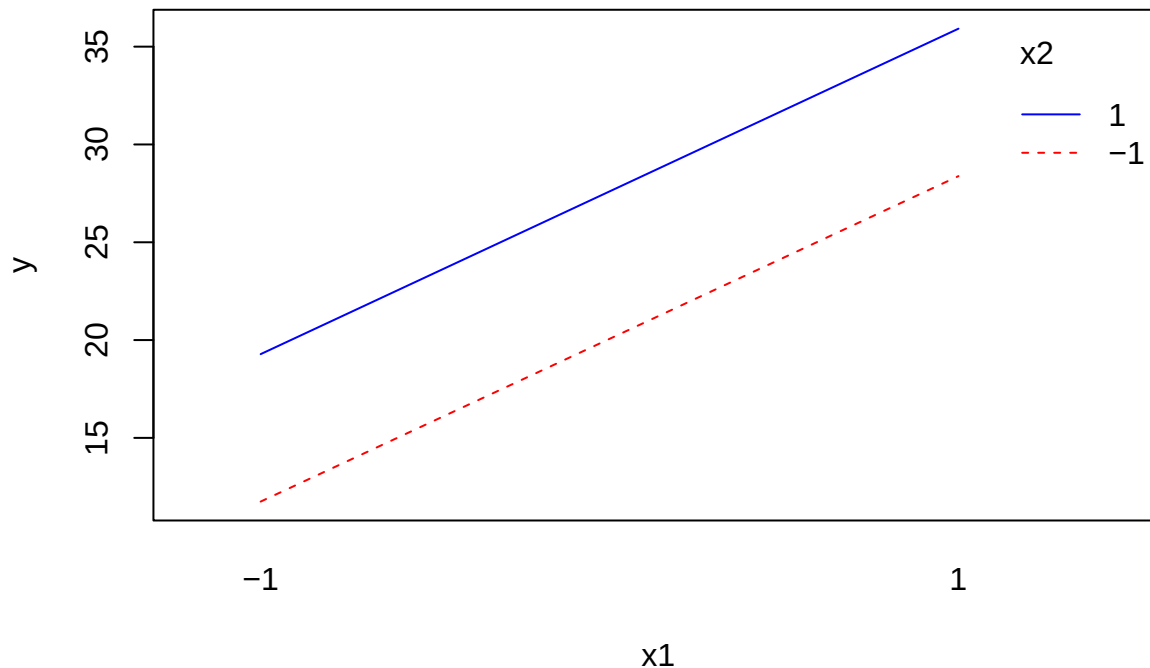
Interaction plot of the interaction model:

```
x.pred = predict(x.mod)
# applying the plot to x1,x2,x.pred same as
# applying the plot to x1,x2,y.
interaction.plot(data$x1,data$x2,x.pred,
                 col=c("red","blue"),
                 trace.label="x2",
                 xlab="x1",
                 ylab="y")
```



Interaction plot of the additive model:

```
add.mod = lm(y~x1+x2,data=data)
add.pred = predict(add.mod)
interaction.plot(data$x1,data$x2,add.pred,
                 col=c("red","blue"),
                 trace.label="x2",
                 xlab="x1",
                 ylab="y")
```



### Part (e)

Test for an interaction effect. (Compute the test statistic and p-value.) Provide an interpretation, stated in the context of the problem.

```
anova(add.mod,x.mod)
```

```
## Analysis of Variance Table
##
## Model 1: y ~ x1 + x2
## Model 2: y ~ x1 * x2
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      13 375.35
## 2      12  71.72  1   303.63 50.801 1.201e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$F^* \approx 50$  is quite large ( $p = .000$ ). We find the data to be incompatible with the additive model.

## Problem 2

A sample of healthy females is selected to investigate the relationship between age ( $x$ ) and the level of a steroid ( $y$ ). Refer to the data from Exercise 8.6.

### Part (a)

Fit a quadratic model using coded variables. Compute  $R^2$  for the quadratic model.

```
data = read.table('CH08PR06.txt')
names(data) = c('y','x')
head(data)
```

```
##      y      x
## 1 27.1 23
```

```
## 2 22.1 19
## 3 21.9 25
## 4 10.7 12
## 5 1.4 8
## 6 18.8 12

# poly uses coded variables (orthogonal)
#      => z1 = a1 + b1 x
#      z2 = a2 + b2 x + c2 x^2
quad.mod = lm(y ~ poly(x,2),data=data)
coef(quad.mod)
```

```
## (Intercept) poly(x, 2)1 poly(x, 2)2
##      17.64444      28.16524     -15.90551
```

```
r2 = summary(quad.mod)$r.squared
r2
```

```
## [1] 0.8143372
```

The estimate for the quadratic regression model is

$$E(Y|x) = 17.644 + 28.165x - 15.906x^2,$$

for which

$$R^2 = 0.8143372.$$

## Part (b)

Fit a cubic model using coded variables. Compute  $R^2$  for the cubic model.

```
cubic.mod = lm(y ~ poly(x,3),data=data)
coef(cubic.mod)

## (Intercept) poly(x, 3)1 poly(x, 3)2 poly(x, 3)3
##      17.644444      28.165236     -15.905513      4.086111

r2 = summary(cubic.mod)$r.squared
```

The estimate for the cubic regression model is

$$E(Y|x) = 17.644 + 28.165x - 15.906x^2 + 4.086x^3,$$

which, given the orthogonal design, has the same coefficients as previously for the lower-order terms.

As expected,  $R^2$  has increased,

$$R^2 = 0.8273324,$$

but only slightly.

## Part (c)

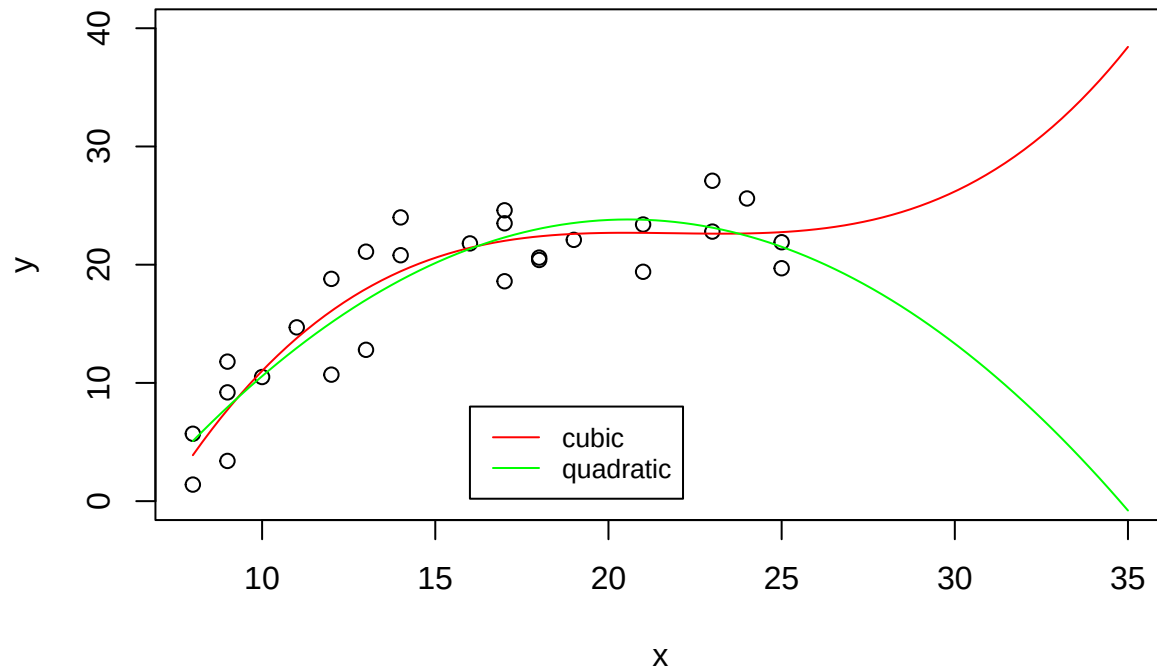
Create a scatterplot of the data, comparing the fitted values from the quadratic model with the fitted values from the cubic model. Extrapolate the predictions 10 years beyond the largest age in the data set.

```
newdat = data.frame(x=seq(min(data$x),max(data$x)+10,length.out=100))
newdat$cubic.pred = predict(cubic.mod, newdata = newdat)
newdat$quad.pred = predict(quad.mod, newdata = newdat)
```

```

plot(data$x,data$y,
      xlim=c(min(data$x),max(data$x)+10),
      ylim=c(0,40),
      xlab="x",
      ylab="y")
legend(16, 8, legend=c("cubic","quadratic"),
      col=c("red","green"), lty=1:1, cex=0.8)
points(newdat$x,newdat$cubic.pred,type="l",col="red")
points(newdat$x,newdat$quad.pred,type="l",col="green")

```



Informally, unless we have prior information (e.g., scientific insight), the simpler (quadratic) model seems like a more likely fit to the underlying data generating process.